Freight Rate Prediction

Machine Learning Engineer assessment - Syed Faizan Ali - November 2026

Summary

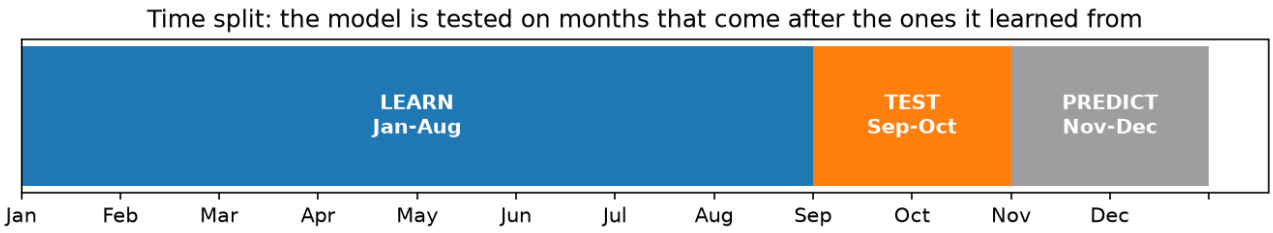
The task is to predict the price of 12,000 truck loads in November-December 2025, using 48,000 loads from January-October 2025 with known prices. The final model is LightGBM using distance, weight, trailer type, pickup and delivery coordinates, and day of week.

Test (learn Jan-Aug, test Sep-Oct)	Average error per load
Simple baseline: median $/mile by trailer type x distance	$229 (10.49%)
Model	**$99 (4.35%)**
Model, normal loads only (without the 1.4% broken prices)	$43 (1.83%)

1. Data problems and fixes

Issue	Training	Validation	Fix
Missing weight	300	165	Fill with the median weight
Negative weight	292	145	Make positive (sign typo: the values otherwise look normal)
Broken prices (outside $1.50-$4 per mile; clear gaps in the histogram)	677 (1.4%)	-	Removed from training only. Validation rows are never removed
quote_signal unreliable (section 3)	half the days	every day	Not used
Cities not seen in training	-	8 cities, 12% of loads	Use coordinates instead of city names

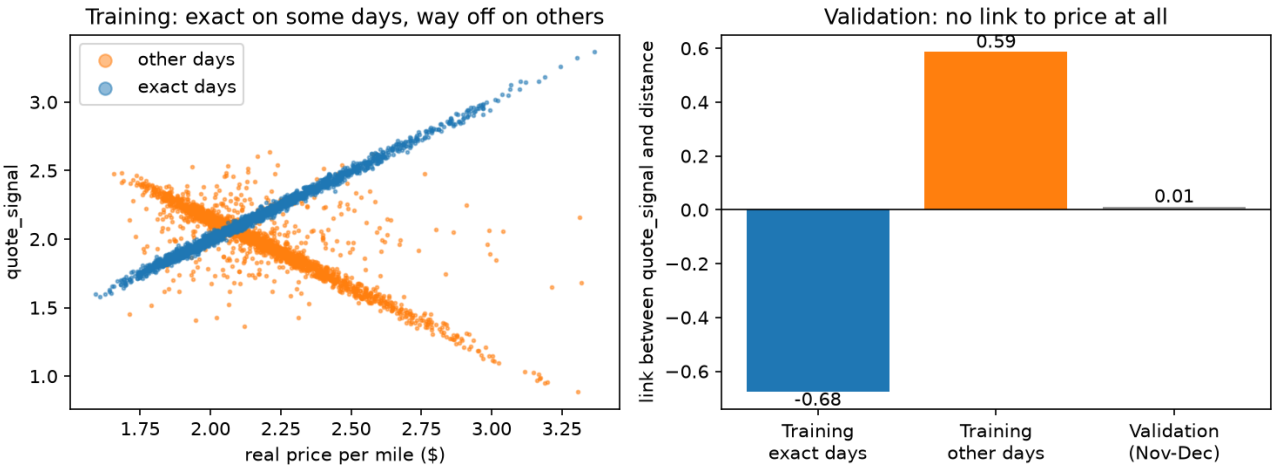
2. Data split and validation



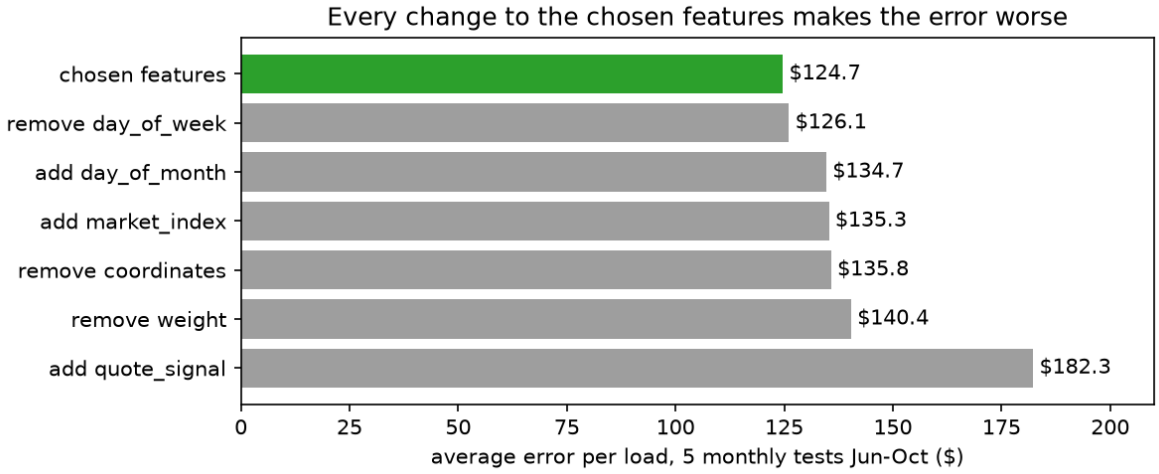
- **Why a time split:** the real task is predicting months the model has never seen (Nov-Dec). So the test copies that: learn from January-August, test on September-October (9,523 loads, all kept, including broken prices).
- **Monthly tests:** to check that results are consistent, each month from June to October is also tested, learning from all earlier months. Every feature choice below is based on these 5 tests.

- **Random split, for comparison:** an 80/20 random split gives $117, close to the time split. The final features have almost nothing tied to a specific day, so mixing months doesn't help the model.
- **Final model:** retrained on all of January-October before predicting November-December.

3. Feature choices

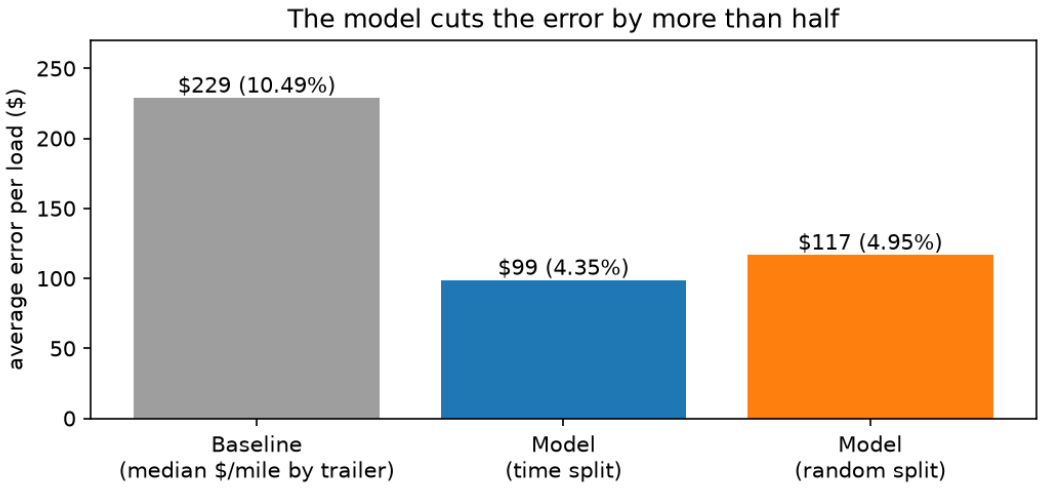


quote_signal looked like the best column but is a trap. On half the training days it equals the real price per mile. On the other half it points the wrong way. In validation it is about 2.05 for every load, with no link to price. A model that trusts it would be badly wrong in November-December, so it is not used.



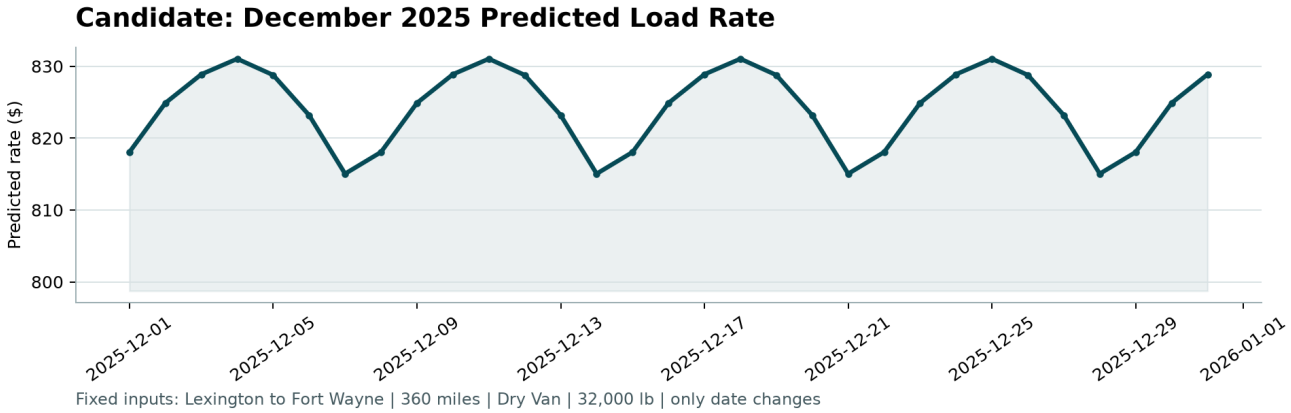
- **market_index** is not used: its link to price drifts during the year, and adding it made the error worse, mostly in September-October.
- **Coordinates instead of city names:** they also work for unseen cities. Hiding 8 cities during training gave $91 error on those loads vs $102 on known ones.
- **LightGBM** was chosen because it is strong on table data, needs no scaling and trains in seconds. It predicts price per mile, which is then multiplied by distance.

4. Results



The model cuts the baseline error by more than half. On normal loads the error is $43 (1.83%). The higher overall average comes from the few loads with broken prices, which no model could predict.

5. December prediction chart



- Lexington to Fort Wayne, 360 miles, Dry Van, 32,000 lb. Coordinates are taken from the training data.
- Predictions range from **$815 (Sundays) to $831 (Thursdays)**, the same midweek pattern seen in training. This is in line with this route's recent prices ($778-$865 in September-October).
- The line repeats each week because day of week is the only input that changes.

6. Limitations

- Error depends on the month: about $100 in August-October, but $212 in June, when prices jumped to a level not seen before. A sudden market change in November-December would not be predicted.
- No holiday effect was found in training, so Thanksgiving and Christmas are treated as normal days.